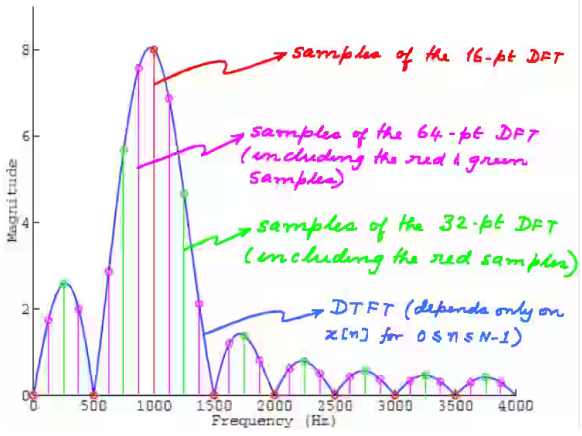
EE2004 Digital Signal Processing

Tutorial 11 Answer Key

1. $X[\cdot] = \{8,\ \frac{1}{2} + j\frac{\sqrt{3}}{2},\ \frac{1}{2} - j\frac{\sqrt{3}}{2}\}$
2. $X[\cdot] = \{3,\ 5 + j2,\ -1,\ 5 - j2\}$
3. (a) $\{1,\ 1 - j2,\ 1\}$
   (b) $\frac{1}{8}\{9,\ -5,\ 1,\ 3,\ 1,\ -5,\ 1,\ 3\}$
4. Computer exercise
5. Computer exercise
6. (a) $\{4,\ 0,\ 0,\ 0\}$
   (b) $\{1,\ 1,\ 1,\ 1,\ 0,\ 0,\ 0,\ 0,\ 0,\ 0\}$
7. (a) $X_a[\cdot] = \{2,\ -j2,\ -2,\ j2\}$
   (b) $X_b[\cdot] = \{4,\ j4,\ -4,\ -j4\}$
   (c) $X_c[\cdot] = \{j,\ -j,\ j,\ -j\}$
   (d) $X_c[\cdot] = \{6 - j,\ -j,\ -6 - j,\ j3\}$
8. $\{-5,\ 11, -5,\ -1\}$
9. $\{1,\ -3,\ 3,\ -1,\ -6,\ 14,\ -8\}$
   For $N = 7$, one period of circular convolution matches the result of circular convolution.
10. 

11. Hint: Suppose the sequence has even symmetry around $n = 0$. Consider the $n < 0$ indices. To what indices in the range 0 to $N - 1$ do they map to?